UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 006 · Problems on Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. Figure G1.1 · The scaling test on log-log axes
2. Figure G4.1 · The extension function $f(u)/f'(u)$ for each choice of f

Colours follow the Okabe–Ito order: blue, vermilion, green, purple for the four deck functions in G1.1; grey dashed lines are references. The deck has no graphs, so both figures are added for clarity and drawn from the deck's own functions.

1 Testing homogeneity: the scaling test

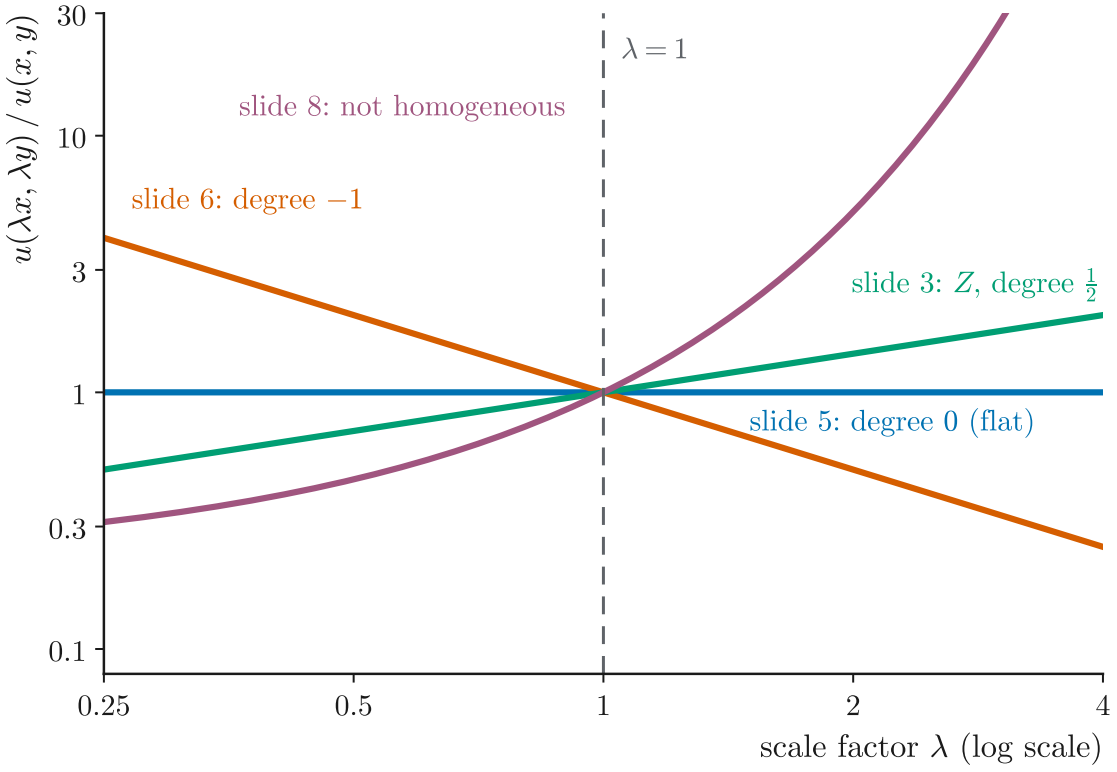


Figure G1.1. The scaling test on log-log axes. ESA ★★

ADDED FOR CLARITY S006·3 · S006·5 · S006·6 · S006·8

How to read it. Each curve is $u(\lambda x, \lambda y)/u(x, y)$ for a deck function, plotted against λ on log-log axes. A homogeneous function gives λ^n: a straight line of slope n through $(1, 1)$. Slide 006·5 is flat (degree 0), slide 006·6 falls with slope -1, Z from slide 006·3 rises with slope $\frac{1}{2}$. Slide 006·8's function curves away: it is not homogeneous. Base point $(1, 2)$; $(2, 1)$ for slide 006·6, where $\sin^{-1}(y/x)$ needs $y \leq x$.

4 When f(u) is homogeneous: the extension

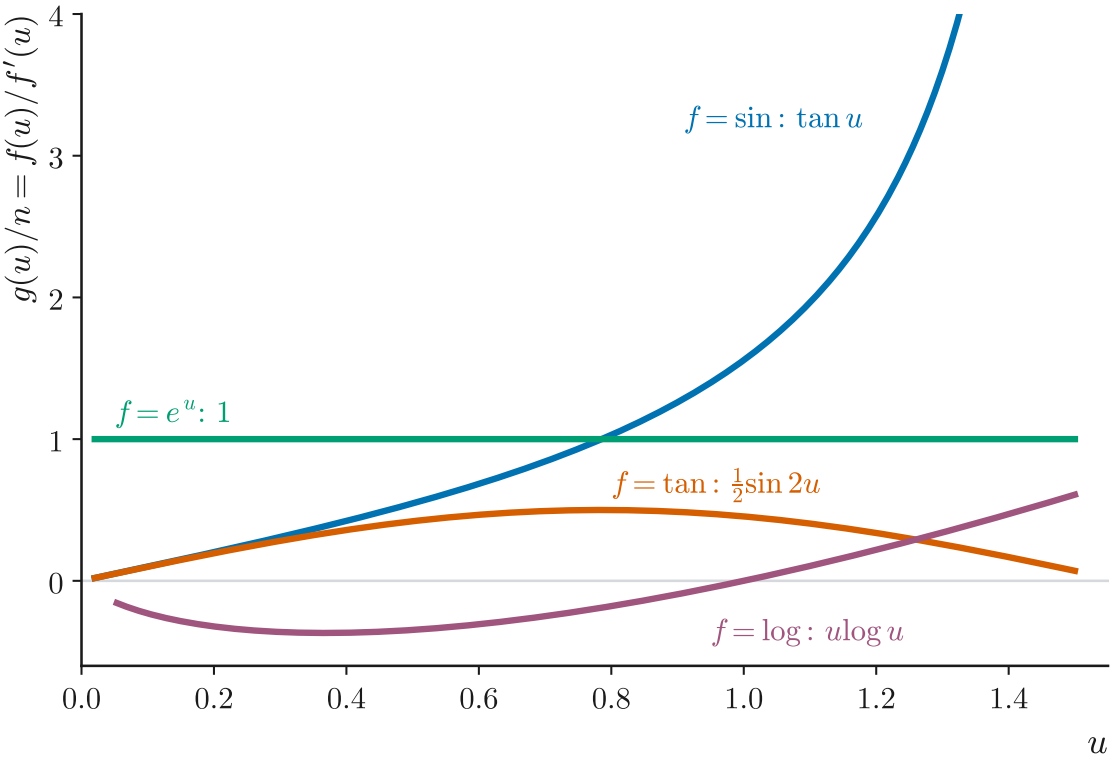


Figure G4.1. The extension function $f(u)/f'(u)$ for each choice of f. ESA ★★

ADDED FOR CLARITY S006·13 · S006·15 · S006·17 · ESA May-24 1a

How to read it. Multiply the curve by the degree n to get $g(u) = xu_x + yu_y$. For $f = \sin$ the sum grows without bound as $u \to \pi/2$; for $f = \tan$ it stays between $\pm\frac{n}{2}$; for $f = e^u$ it is the constant n, so the second order is $-n$; for $f = \log$ it is $nu\log u$, negative for $u < 1$ (on slide 006·12 $u = e^{(\cdots)} > 1$, so it is positive).